CS-600: Software Design and Development

6-1 Short Paper: Feature Expansion Planning

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Development Plan for Video Game Guru Application

Technology applications must continue improving to stay competitive and meet the changing needs of users. If software systems are not updated regularly, customers may begin using competing applications that provide better tools and features. The Video Game Guru application was originally created to help distributors, retailers, and advanced gaming users analyze information about video game studios, genres, game titles, and sales performance. The next phase of development focuses on improving the application by adding new features requested by users while also supporting long-term business growth. The core business goals for this project are to modernize the application, increase customer retention, and attract new customers by offering more advanced functionality. The company hopes these improvements will increase the customer base by 35% and improve yearly revenue from maintenance subscriptions. To support these goals, this development plan finds new feature opportunities, explains updated requirements, evaluates sustainability concerns, and justifies the proposed design choices.

New Feature Opportunities

One important feature opportunity is adding machine requirement information for each game title. Users requested the ability to view operating system compatibility, CPU requirements, RAM requirements, storage requirements, and networking specifications for games. This feature would help users compare games more effectively and name which studios are using the newest technologies within different genres. It would also allow distributors and retailers to better understand the technical demands of modern games.

Another feature request involves adding rating functionality for video game titles. Users want to see both official age ratings and player review scores for games. This feature would improve the user experience because customers could quickly evaluate games based on community feedback and recommended age groups. User ratings would also increase interaction within the application because users could take part by reviewing titles they have played. A third feature opportunity focuses on improving the existing sales reporting system. The application already stores total sales information, but users requested advanced sorting and filtering tools that organize sales data by genre and studio. This would help companies analyze market trends and show successful development partnerships more efficiently. Together, these new features would modernize the application and make it more useful for both current and future customers.

Updated and Supporting Requirements

Several updated requirements are necessary to support the new features. For the machine requirement feature, the application shall store operating system compatibility, minimum and recommended CPU requirements, RAM requirements, storage space requirements, and networking information for each title. Users shall also be able to filter and sort games by genre, studio, and hardware requirements. To support game ratings, the application shall support official age ratings like ESRB or PEGI. The system shall also allow registered users to send player ratings using a five-star scale. The application shall calculate and display average user ratings for each game title. To keep data accuracy, the system shall prevent users from giving duplicate ratings for the same game. For the enhanced sales analytics feature, the application shall allow users to sort and compare sales data by genre, studio, title popularity, and release year. The application shall also provide charts and visual reports within the web-based user interface to improve readability and help users analyze trends more effectively. Other supporting requirements are also needed. Since users will now create ratings and reviews, the application must support secure authentication and account management. Database storage capacity must also increase to handle other metadata and user-generated content. Existing role-based access control should continue protecting administrative functions and sensitive data. The system shall continue visualizing relationships between studios, genres, titles, and sales of information through the user interface.

Sustainability and Scalability Considerations

Sustainability is an important part of software development because applications must continue functioning effectively as the number of users and features grows over time. One major consideration for this project is database scalability. The application will now store significantly more information, including hardware specifications, ratings, and detailed analytics. A scalable database structure and optimized indexing methods will help keep good system performance as the application grows. Maintainability is another important concern. The application should follow a modular architecture, so developers can update features independently without affecting the entire system. Modular development improves flexibility, reduces technical debt, and makes future updates easier to implement. According to Pressman and Maxim (2020), designing software with maintainability and scalability in mind helps organizations reduce long-term development costs and improve software quality. Performance optimization is also necessary because advanced filtering and analytics queries may increase processing demands. Chandrasekaran and Krishnan (2022) explain that purpose-built query models and optimized query systems improve scalability and allow systems to adapt more efficiently to changing business needs. Using optimized query structures and caching techniques would improve application performance and user experience. Security is another important sustainability factor. Since the application now includes user accounts and reviews, strong authentication and encryption methods are necessary to protect user information. Proper validation controls will also help prevent spam, duplicate reviews, and unauthorized access to administrative features. The development team should also prepare for future expansion. Added features such as recommendation systems, social features, or AI-based analytics could eventually be added to the platform. Designing flexible APIs and scalable backend services will now make future development easier and more cost-effective.

Justification of Design Choices

The proposed features directly support both user feedback and business goals. Adding machine requirement information improves the platform’s usefulness for advanced gaming users, distributors, and retailers by providing detailed technical information that competing applications may not offer. The rating system improves customer engagement because users can actively take part in reviewing games and sharing feedback. Applications that encourage user interaction often create stronger customer loyalty and improve user retention. The enhanced analytics and reporting tools support the company’s business goals by helping users show profitable trends, successful game studios, and market opportunities. Pressman and Maxim (2020) note that software features should align with business goals and user needs to maximize value and support organizational success. These tools make the application more valuable for professional users who rely on data analysis for business decisions.

The decision to use a scalable and modular system architecture supports long-term sustainability. Chandrasekaran and Krishnan (2022) explain that query-driven architectures and event-based systems make it easier for applications to evolve as requirements change over time. This approach reduces maintenance complexity and supports future system growth. The proposed design choices improve usability, increase competitiveness, and support the long-term success of the Video Game Guru application.

Conclusion

The Video Game Guru application can significantly improve its value and competitiveness by adding machine requirement tracking, game ratings, and advanced sales analytics features. These improvements align with both user feedback and business aims while creating a more modern and engaging platform. This development plan also considers long-term sustainability by addressing scalability, maintainability, performance, and security concerns. By using flexible and scalable software design practices, the application will be better prepared for future growth and more feature expansion. The proposed features not only address user feedback but also support software engineering best practices related to scalability, maintainability, and long-term product evolution (Pressman & Maxim, 2020).

References

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